AI AND AUTOMATION PROGRAM · ADVANCED LEVEL

15 Capstone Project Ideas

End-to-end AI automation systems built across Modules 6 to 10

How to use this document

Each participant (or team) chooses **one** idea as their capstone, or adapts one to their own domain. Every idea is a realistic system that companies pay automation engineers to build, and each is designed to pass through the three capstone milestones (Sessions 38 to 40): a first AI workflow, an agent deployed to a real channel, and a fully integrated, monitored and ethically governed system.

#	Project	Domain	Complexity
1	Smart Accounts-Payable Automation	Finance / Back office	●●●●○
2	AI Recruitment Pipeline (Screen, Rank, Schedule)	HR / Talent	●●●●●
3	University Course Assistant (RAG Chatbot)	Education	●●●●○
4	Customer Support Triage & Auto-Reply Agent	Customer service	●●●●○
5	E-commerce Order & Inventory Operations Hub	E-commerce / Retail	●●●●○
6	AI Content Studio with Editorial Approval	Marketing	●●●●○
7	Meeting Intelligence Assistant	Productivity / Operations	●●●●○
8	Contract Review & Risk Flagging Assistant	Legal / Procurement	●●●●●
9	Market & Competitor Intelligence Briefing	Strategy / Business intelligence	●●●●○
10	B2B Lead Generation & Outreach Engine	Sales	●●●●○
11	Clinic Booking & Patient Follow-up System	Healthcare services	●●●●○
12	Exam & Quiz Generator for Educators	Education	●●●●○
13	Real-Estate Listing Matcher	Real estate	●●●●○
14	Talk-to-Your-Data Analyst Agent	Data & analytics	●●●●●
15	IT Incident Response Automation	IT operations / DevOps	●●●●●

What every capstone must include

Whichever idea is chosen, the final system must show the skills from the whole program. The right column points to the session where each skill is taught.

Area	Requirement	Session
Automation foundations	Working API and webhook integrations, authenticated connections to business tools, clean handling of JSON and lists of items.	S16–S20
Prompting	Prompts written with structure, examples and a required output format, collected in the system's Prompt Library.	S7–S11
Workflow design	Branching, loops and routing, sub-workflows for reusable parts, and fallback paths when a step fails.	S21
AI in the workflow	AI used for real decisions or transformations, with the model choice justified (quality vs cost).	S22
Knowledge or documents	A RAG knowledge base grounded in trusted data, **or** real document processing (PDFs, scans, images, audio).	S23, S25
Multi-agent design	Cooperating agents or an orchestrator with specialist agents, where it adds real value.	S24, S29
Events, schedules, files	At least one event-driven trigger and one scheduled job; the system generates a file or formatted report.	S26
Validation & recovery	Schema checks on AI output, retries with a corrective prompt, error workflow, human escalation for risky actions.	S27
Deployed agent	An agent with tools and memory, deployed to Slack, web chat or email and used by real or test users.	S29, S30
Agent evaluation	A test set with quality metrics (correctness, helpfulness, safe refusals) and results before/after changes.	S31
Monitoring & cost	Execution logs, an audit trail, alerts on failure and a simple cost dashboard per run or per day.	S32
Security & privacy	Credentials only in n8n credentials, least privilege, masked personal data, prompt-injection defences.	S33
Ethics guardrails	Three explicit, **testable** ethics guardrails, demonstrated live in the final review.	S34
Governance	Versioned workflows and a separate test (dev) setup before changes reach the live system.	S35
Business case	Who uses it, what it saves, how it will be adopted: a one-page business case and adoption plan.	S36, S37

Suggested stack

n8n · Groq for chat models · Gemini for embeddings and file/audio reading · Supabase/Postgres, Qdrant or Pinecone for storage and vectors · Google Workspace for files, mail and calendars · Slack, web chat (n8n Chat Trigger) or email as the agent channel. Other providers are fine if the choice is justified.

Milestones and deliverables

The capstone runs in parallel with Modules 6 to 8 and is reviewed live three times. Every project page in this document shows what to build for each milestone.

Milestone	When	What is reviewed
#1 Blueprint & first AI workflow	Session 38, after Module 6	Problem statement, target users, one-page architecture diagram, and a live demo of the core workflow ("How it works" on each project page) with at least one AI step and one validation step.
#2 Deployed agent on a real channel	Session 39, after Module 7	The project's agent ("Deployed agent" on each page) live on Slack, web chat or email, showing its knowledge sources, tools, memory and at least one validation/recovery path.
#3 Final integration, demo & review	Session 40, after Modules 8–9	The full system with monitoring, cost dashboard and security review, the 3 ethics guardrails in action, and the business case and adoption plan. 15 to 20 minute presentation plus Q&A.

Required deliverables

1. System Blueprint	Architecture diagram (data sources, AI tools, n8n workflows, agents, channels, outputs), the domain problem and value, and a phased roadmap: pilot, scaling, governance, training, adoption metrics.
2. Working AI Workflow & Agent	At least one n8n workflow with AI (RAG, multi-agent or document processing as appropriate) plus monitoring, security and error handling, and one agent deployed to a real channel.
3. Prompt, Operations & Ethics Pack	(a) The tested Prompt Library used by the system; (b) a runbook for monitoring, retries, fallback and human escalation; (c) a security and ethics review covering credentials, sensitive data, prompt injection and the 3 guardrails.
4. Final Presentation & Live Demo	15 to 20 minutes end to end, a live demo of at least one working component, the 3 guardrails shown in action, followed by Q&A.

Weight in the final grade

The Final Capstone Project (three milestones plus final demo) is worth **40%** of the program grade. Passing requires at least 60% overall **and** a capstone submitted across all three milestones; 85% or more earns a Distinction.

Reading the project pages

Complexity runs from ●●●●○ (advanced) to ●●●●● (most demanding). Each page lists the problem, the core workflow, the deployed agent, three example ethics guardrails, the sessions it draws on most, suggested n8n nodes, success criteria and a stretch goal. The guardrails are examples: participants define and test their own in Session 34.

01 Smart Accounts-Payable Automation

Finance / Back office · Complexity ●●●●○

Turns supplier invoices arriving by email into validated accounting entries, with a human approval step for anything unusual.

THE PROBLEM

Finance teams retype hundreds of invoices a month from PDFs and photos into spreadsheets or ERP systems. It is slow, error-prone and a common first target for automation consultants.

HOW IT WORKS · MILESTONE #1

1. Gmail trigger watches an **invoices@** label; attachments are split into one item per file.
2. Gemini reads each PDF or photo and returns **structured JSON** (supplier, invoice number, date, line items, VAT, total).
3. A validation layer checks the maths (line items add up to total), required fields and duplicate invoice numbers; failures retry with a corrective prompt.
4. Invoices above a threshold or from new suppliers go to a **Wait** node with a Slack approve/reject button.
5. Approved rows are written to Google Sheets (or Airtable) and the original file is archived to a Drive folder per month.
6. A scheduled workflow sends a weekly spend summary PDF to the finance manager.

DEPLOYED AGENT · MILESTONE #2

A finance assistant in **Slack**: "What do we owe supplier X this month?" or "Show unpaid invoices over 50,000 EGP", answered from the invoice sheet, with memory for follow-ups.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The AI never approves or pays an invoice; anything above the threshold or from a new supplier needs a named human approver.
- Supplier bank details are masked before any text is sent to the model.
- Every extracted value can be traced back to the original file, which is kept unchanged in the archive.

SESSIONS IT DRAWS ON MOST

S21 Intermediate workflows · S22 AI in automations · S25 Files & OCR · S26 File generation & scheduling · S27 Validation & recovery · S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Gmail Trigger · Google Gemini (document) · Structured Output Parser · Code · IF / Switch · Wait (approval) · Google Sheets · Google Drive · Schedule Trigger · Error Trigger

SUCCESS CRITERIA

- Processes a batch of 20 mixed invoices (clean PDFs, scans, phone photos) with an accuracy log.
- Duplicate and maths errors are caught, never written silently.

STRETCH GOAL

Detect currency and convert EGP/USD with a live exchange-rate API; flag invoices whose prices differ from the last order by more than 15%.

02 AI Recruitment Pipeline (Screen, Rank, Schedule)

HR / Talent · Complexity ●●●●●

Scales the Session 24 Job Application Coach idea to the recruiter's side: screens every applicant for a role and books interviews for the shortlist.

THE PROBLEM

Recruiters receive hundreds of CVs per opening and spend most of their time on first-pass screening. Companies actively hire people who can automate this fairly and transparently.

HOW IT WORKS · MILESTONE #1

1. HR publishes a job through an n8n Form; applicants apply through a second form (or a Gmail label) with their CV.
2. Each CV is transcribed, then a multi-agent team runs: **Requirements Extractor**, **CV Scorer** (scored rubric per requirement), **Bias Checker** (flags reasons that use age, gender or nationality).
3. Scores are validated as JSON with a fixed schema; inconsistent scores trigger a second-opinion agent.
4. A ranked shortlist is written to Google Sheets with the reasoning per candidate.
5. Top candidates receive an email with Google Calendar slots; rejected candidates receive a polite, personalised email after HR approval.

DEPLOYED AGENT · MILESTONE #2

A recruiter assistant in **Slack** ("Why is candidate X ranked third?") plus an **email** agent that answers applicants' questions about the role and process.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Age, gender, religion, nationality and photos are removed before scoring, and the bias checker blocks any reasoning that uses them.
- No candidate is rejected automatically; every decision email requires HR approval.
- Applicants are told that AI assists the first screening and that a person makes the final decision.

SESSIONS IT DRAWS ON MOST

S22 AI in automations | S24 Multi-agent | S25 Files & OCR | S27 Validation & recovery | S29 Agent patterns & memory | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form Trigger | AI Agent + Agent Tools | Groq / Gemini | Structured Output Parser | Loop Over Items | Google Sheets | Google Calendar | Gmail | Wait

SUCCESS CRITERIA

- Ranks 30 sample CVs for one job with explainable scores.
- No email goes to candidates without an HR approval step.

STRETCH GOAL

A small HR dashboard (n8n Form + Sheets or a webhook-served HTML page) showing pipeline stages and average time-to-shortlist.

03 University Course Assistant (RAG Chatbot)

Education · Complexity ●●●●○

A chat assistant that answers students' questions from the official course material, with citations, and tells the lecturer what students struggle with.

THE PROBLEM

Lecturers answer the same questions repeatedly, and general chatbots answer from outside the syllabus. Universities and e-learning platforms want grounded assistants that cite their own material.

HOW IT WORKS · MILESTONE #1

1. An ingestion workflow watches a Google Drive folder; new lecture PDFs and slides are chunked, embedded with Gemini and stored in a vector store.
2. Students ask on the course web chat or Slack; an AI Agent retrieves the top chunks and answers **only** from them, citing lecture and page.
3. Low-confidence or out-of-scope questions get an honest "not in the material" reply and are logged.
4. Conversation memory keeps context per student for follow-up questions.
5. A weekly scheduled workflow clusters logged questions and emails the lecturer a "topics students struggle with" report.

DEPLOYED AGENT · MILESTONE #2

A course assistant on **web chat** (n8n Chat Trigger embedded in the course page) or **Slack**, with per-student memory for follow-up questions.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Answers only from the uploaded course material, always with a citation; otherwise it says it does not know.
- It explains concepts but refuses to write full solutions to graded assignments.
- Students are stored under a pseudonymous ID; question logs contain no names or emails.

SESSIONS IT DRAWS ON MOST

S21 Intermediate workflows · S23 RAG · S25 Files & OCR · S26 File generation & scheduling · S29 Agent patterns & memory · S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Google Drive Trigger · Default Data Loader · Text Splitter · Embeddings Gemini · Vector Store (Supabase / Qdrant / Pinecone) · AI Agent · Window Buffer Memory · Chat Trigger (web chat) · Schedule Trigger

SUCCESS CRITERIA

- Answers a 25-question test set; at least 85% grounded and correctly cited.
- Re-ingests a changed file without duplicating chunks.

STRETCH GOAL

Bilingual answers (Arabic question, English source) and a "quiz me" command that generates practice questions from a chosen lecture.

04 Customer Support Triage & Auto-Reply Agent

Customer service · Complexity ●●●●○

Classifies incoming tickets, drafts grounded replies from the help centre, and escalates to a human when confidence is low or an SLA is about to break.

THE PROBLEM

Support teams drown in repetitive tickets while urgent ones wait. AI-assisted triage is one of the most requested automation services for SMEs and SaaS companies.

HOW IT WORKS · MILESTONE #1

1. Tickets arrive by email, a website form or web chat and are normalised into one schema.
2. A classifier returns category, urgency, sentiment and language as validated JSON.
3. A RAG agent drafts an answer from the FAQ/help-centre knowledge base with a self-reported confidence.
4. High confidence on low-risk categories is sent automatically; everything else becomes a draft for an agent in Slack with approve/edit buttons.
5. A scheduled check every 30 minutes escalates tickets approaching their SLA and posts a daily ticket summary.

DEPLOYED AGENT · MILESTONE #2

A support agent on **email** and **web chat** that answers customers from the help centre, plus a **Slack** channel where staff approve or edit drafts.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The AI never promises refunds, discounts or compensation; those topics always go to a human.
- Instructions written inside a ticket (prompt injection) are treated as customer text, never as commands.
- Phone numbers, card numbers and addresses are masked before the model sees the ticket.

SESSIONS IT DRAWS ON MOST

S21 Intermediate workflows | S22 AI in automations | S23 RAG | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Gmail / Webhook / Chat Trigger | Text Classifier | Information Extractor | Vector Store | AI Agent | Switch | Slack | Google Sheets / Postgres | Schedule Trigger

SUCCESS CRITERIA

- Correct category on at least 90% of a 40-ticket test set.
- A clear audit log: which replies were automatic, which were human-approved.

STRETCH GOAL

Detect angry customers and route them to a senior agent; measure average first-response time before vs after automation.

05 E-commerce Order & Inventory Operations Hub

E-commerce / Retail · Complexity ●●●●○

Event-driven back office for an online store: order processing, stock alerts, supplier purchase orders and daily sales reporting.

THE PROBLEM

Small online stores in Egypt and the region run operations manually across WooCommerce/Shopify, WhatsApp and spreadsheets. n8n freelancers are hired exactly for this kind of glue work.

HOW IT WORKS · MILESTONE #1

1. A webhook receives new orders (from WooCommerce/Shopify or a mock store built with an n8n Form).
2. Stock is decremented in a database; orders with missing data or impossible quantities are rejected with a clear reason.
3. When stock drops below a reorder level, an agent drafts a supplier purchase order, n8n generates it as a PDF and emails it after manager approval.
4. Customers receive order confirmation and shipping updates.
5. A nightly workflow produces a sales report (best sellers, revenue, low stock) as a PDF/Excel file and posts it to Slack.

DEPLOYED AGENT · MILESTONE #2

An operations assistant in **Slack**: "How many units of X are left?", "Draft a purchase order for supplier Y", answered from the database with tools.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The AI can only read stock and draft documents; it cannot change prices or stock levels directly.
- Purchase orders are sent to suppliers only after a manager approves them.
- Customer personal data is never included in AI prompts or in reports.

SESSIONS IT DRAWS ON MOST

S17 APIs & webhooks | S21 Intermediate workflows | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels | S32 Monitoring & cost

SUGGESTED N8N NODES AND SERVICES

Webhook | Postgres / Google Sheets | IF / Switch / Merge | Code | HTML to PDF / Convert to File | Gmail | Chat Trigger (web chat) | Schedule Trigger | Error Trigger

SUCCESS CRITERIA

- Handles 100 simulated orders, including invalid ones, without corrupting stock levels.
- An error workflow notifies the owner on any failure with the order ID.

STRETCH GOAL

Demand forecasting: an AI step reads the last 60 days of sales and suggests reorder quantities with a short justification.

06 AI Content Studio with Editorial Approval

Marketing · Complexity ●●●●○

A multi-agent content team (researcher, writer, editor, designer brief) that produces social posts and blog drafts, waits for approval, then publishes on schedule.

THE PROBLEM

Marketing teams want AI speed without losing brand voice or publishing mistakes. A controlled pipeline with review gates is more valuable than a single prompt.

HOW IT WORKS · MILESTONE #1

1. A content calendar in Google Sheets lists topics, platforms and publish dates.
2. An orchestrator agent calls a **Researcher** (web search tool), a **Writer** (per-platform format), an **Editor** (brand-voice checklist and fact-check) and an **Image Prompt** agent.
3. Output is validated: character limits per platform, required hashtags, banned words.
4. The package goes to the marketing lead by email/Slack with approve, request-changes or reject; change requests loop back to the Writer.
5. Approved posts are published at the scheduled time (LinkedIn/X/Facebook or a mock endpoint) and logged back to the sheet.

DEPLOYED AGENT · MILESTONE #2

A content assistant in **Slack** where the team requests new posts, reviews drafts in threads and approves them with buttons.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Nothing is published without a named human approval.
- Facts and statistics must come with a source link; the editor agent rejects unsupported claims.
- No copyrighted text or images are reused, and AI-generated content is labelled where platform rules require it.

SESSIONS IT DRAWS ON MOST

S22 AI in automations | S24 Multi-agent | S26 File generation & scheduling | S27 Validation & recovery | S29 Agent patterns & memory | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Google Sheets | AI Agent + Agent Tools | HTTP Request (search API) | Structured Output Parser | Wait (approval) | Schedule Trigger | LinkedIn / HTTP Request | Slack

SUCCESS CRITERIA

- A full week of content for one brand generated and approved end to end.
- The revision loop is capped (e.g. 2 rounds) to avoid infinite loops.

STRETCH GOAL

Engagement feedback: a weekly job reads post metrics and tells the Writer agent which styles performed best.

07 Meeting Intelligence Assistant

Productivity / Operations · Complexity ●●●●○

From a meeting recording to a summary, decisions, action items in the team's task board and follow-up emails, automatically.

THE PROBLEM

Action items from meetings are forgotten or never assigned. Every company runs meetings, so this is easy to demo and easy to sell.

HOW IT WORKS · MILESTONE #1

1. A recording is uploaded through a form or dropped into a Drive folder.
2. Audio is transcribed (Groq Whisper or Gemini audio); long files are split and merged.
3. Agents produce an executive summary, a decisions list and action items as validated JSON (owner, task, due date).
4. Action items are created in Trello/Notion/ClickUp and assigned; unknown owners are flagged instead of guessed.
5. A formatted minutes document (PDF or Google Doc) is generated and emailed to attendees; a reminder runs the day before each due date.

DEPLOYED AGENT · MILESTONE #2

A **Slack** bot that answers questions about past meetings ("What did we decide about the budget in March?") using RAG over transcripts.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Recordings are processed only after attendees are told (a consent notice is part of the invite).
- Owners and due dates are never invented; missing ones are marked for review.
- Meetings marked confidential are excluded, and raw audio is deleted after transcription.

SESSIONS IT DRAWS ON MOST

S22 AI in automations | S23 RAG | S25 Files & OCR | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form / Google Drive Trigger | HTTP Request (Groq Whisper) | Google Gemini | Information Extractor | Trello / Notion | Google Docs | Gmail | Schedule Trigger

SUCCESS CRITERIA

- Works on a 30-minute recording within a few minutes.
- Owners and dates are never invented; missing ones are marked for review.

STRETCH GOAL

Ask-the-archive: a RAG chatbot over all past meeting transcripts ("what did we decide about the budget in March?").

08 Contract Review & Risk Flagging Assistant

Legal / Procurement · Complexity ●●●●●

Reads a contract, extracts key clauses, compares them with the company's legal checklist and produces a risk report for the reviewer.

THE PROBLEM

Legal and procurement teams review many similar contracts (NDAs, service agreements, leases). First-pass review against a checklist is repetitive and well suited to RAG plus structured extraction.

HOW IT WORKS · MILESTONE #1

1. A contract PDF or DOCX is uploaded; scanned contracts go through OCR.
2. An extractor pulls parties, dates, term, payment terms, termination, liability, confidentiality and governing law into a fixed schema.
3. A RAG agent compares each clause with the company playbook (stored in a vector store) and rates it green / amber / red with a reason.
4. A second agent cross-checks red flags to reduce false alarms.
5. A risk report (PDF/DOCX) with a summary table and suggested wording is generated and emailed; results are logged per contract.

DEPLOYED AGENT · MILESTONE #2

An **email** agent: send a contract to a review address and get the risk report back, plus **Slack** Q&A about clauses in reviewed contracts.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Every report is labelled "first-pass review, not legal advice" and needs a reviewer's sign-off.
- Each flag quotes the exact clause it is based on, so nothing is asserted without evidence.
- Confidential contracts go only to model providers whose terms exclude training on submitted data.

SESSIONS IT DRAWS ON MOST

S23 RAG | S24 Multi-agent | S25 Files & OCR | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form Trigger | Extract from File / Gemini OCR | Information Extractor | Vector Store | AI Agent + Agent Tools | Code | Convert to File / HTML to PDF | Gmail

SUCCESS CRITERIA

- Tested on 10 real or public sample contracts with a reviewer's verdict for comparison.
- Every flag points to the exact clause text it came from.

STRETCH GOAL

Version comparison: upload two versions of the same contract and get a list of changed clauses with risk impact.

09 Market & Competitor Intelligence Briefing

Strategy / Business intelligence · Complexity ●●●●○

A scheduled research system that monitors news, competitor sites and job posts, removes noise, and delivers a weekly executive briefing.

THE PROBLEM

Managers want to know what competitors are doing but nobody has time to read everything. Automated monitoring with AI summarisation is a common consulting deliverable.

HOW IT WORKS · MILESTONE #1

1. A daily schedule pulls RSS feeds, news APIs and selected web pages for a list of competitors and keywords.
2. Items are de-duplicated (hash and similarity check) and stored in a database.
3. An agent tags each item (product launch, pricing, hiring, partnership) and scores its importance.
4. Important items trigger an instant Slack alert.
5. Every Sunday, a briefing PDF with sections per competitor and "what this means for us" is generated and emailed.

DEPLOYED AGENT · MILESTONE #2

A **Slack** agent that answers questions over the collected archive ("What has competitor X done on pricing this quarter?").

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Only public sources whose terms allow automated collection are used, and this is documented per source.
- No personal data about individual employees of competitors is collected or stored.
- Summaries link to the original source instead of copying full articles.

SESSIONS IT DRAWS ON MOST

S17 APIs & webhooks | S21 Intermediate workflows | S22 AI in automations | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Schedule Trigger | RSS Read | HTTP Request | HTML Extract | Remove Duplicates | Postgres / Supabase | AI Agent | HTML to PDF | Gmail / Slack

SUCCESS CRITERIA

- Runs unattended for at least one week before the final demo.
- Duplicate stories from different sources appear once.

STRETCH GOAL

A RAG chat over the collected archive so managers can ask "what has competitor X done on pricing this quarter?".

10 B2B Lead Generation & Outreach Engine

Sales · Complexity ●●●●○

Captures leads, enriches and scores them, writes personalised outreach, runs a follow-up sequence and keeps the CRM up to date.

THE PROBLEM

Sales teams lose leads through slow follow-up and generic emails. Lead automation is one of the highest-paid n8n freelance categories.

HOW IT WORKS · MILESTONE #1

1. Leads come from a website form, a CSV import or a webhook from an ads platform.
2. Enrichment: company website is fetched and summarised; the lead is scored against an ideal-customer profile as validated JSON.
3. Hot leads go to a salesperson immediately; warm leads receive a personalised first email written by an agent and checked by a reviewer agent.
4. A sequence with Wait nodes sends up to two follow-ups and stops automatically when the lead replies.
5. All activity is synced to a CRM (HubSpot free tier, Airtable or Sheets) and a weekly pipeline report is sent.

DEPLOYED AGENT · MILESTONE #2

An **email** agent that reads replies, classifies them (interested, not now, unsubscribe, question) and answers simple questions, with a **Slack** hand-off to sales.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Unsubscribe requests are honoured immediately and the address is never contacted again.
- Personalisation uses only facts found on the lead's public company pages; nothing is invented.
- Daily sending limits and a clear sender identity are enforced in every email.

SESSIONS IT DRAWS ON MOST

S17 APIs & webhooks | S22 AI in automations | S24 Multi-agent | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form / Webhook | HTTP Request | AI Agent | Structured Output Parser | Wait | Gmail Trigger (reply detection) | HubSpot / Airtable | Schedule Trigger

SUCCESS CRITERIA

- A full sequence demonstrated on test inboxes, including the stop-on-reply rule.
- Opt-out handling: unsubscribed leads are never emailed again.

STRETCH GOAL

A/B test two email styles and report which one gets more replies.

11 Clinic Booking & Patient Follow-up System

Healthcare services · Complexity ●●●●○

A messaging assistant that books, reschedules and reminds patients, then collects feedback after the visit.

THE PROBLEM

Clinics in Egypt manage bookings by phone and WhatsApp, with many no-shows. A booking assistant with reminders has clear, measurable value.

HOW IT WORKS · MILESTONE #1

1. Patients chat with the clinic's web chat assistant; an agent understands the request (book, cancel, reschedule, ask about prices or hours).
2. The agent uses tools to check doctor availability in Google Calendar and to create or move appointments.
3. Clinic FAQs (prices, location, preparation instructions) are answered from a small knowledge base.
4. Scheduled reminders go out 24 hours and 2 hours before each appointment; no reply triggers a confirmation request.
5. After the visit, a feedback message is sent and responses are summarised weekly for the clinic manager.

DEPLOYED AGENT · MILESTONE #2

A booking agent on the clinic's **web chat**, with **email** confirmations and reminders (WhatsApp can be added as an extra channel).

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The assistant never gives medical advice or interprets symptoms; it hands over to clinic staff.
- Urgent messages ("chest pain", "bleeding") immediately show the emergency number and alert staff.
- Patient data is limited to what booking needs and is visible only to clinic staff.

SESSIONS IT DRAWS ON MOST

S21 Intermediate workflows · S22 AI in automations · S23 RAG · S26 File generation & scheduling · S29 Agent patterns & memory · S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Chat Trigger (web chat) · AI Agent · Google Calendar Tool · Window Buffer Memory · Vector Store · Schedule Trigger · Google Sheets · Error Trigger

SUCCESS CRITERIA

- Handles double-booking attempts and invalid dates safely.
- The bot never gives medical advice; it hands over to clinic staff.

STRETCH GOAL

Waiting-list automation: when someone cancels, the first person on the waiting list is offered the slot automatically.

12 Exam & Quiz Generator for Educators

Education · Complexity ●●●●○

Turns lecture material into a reviewed question bank, a ready-to-use online quiz and a printable answer key.

THE PROBLEM

Writing good exam questions takes lecturers many hours, and AI-generated questions often contain wrong answers. A pipeline with a reviewer agent and validation solves the quality problem.

HOW IT WORKS · MILESTONE #1

1. A lecturer uploads lecture PDFs and chooses the number of questions, difficulty mix and question types.
2. A **Question Writer** agent creates questions with answers and explanations, grounded in the uploaded text.
3. A **Reviewer** agent checks each question against the source: one correct answer, plausible distractors, no ambiguity. Rejected questions are rewritten.
4. Questions are validated as JSON and saved to a question bank (Sheets or a database) tagged by topic and difficulty.
5. n8n generates a Google Form quiz (or a webhook-served quiz page) and a PDF answer key, and emails both to the lecturer.

DEPLOYED AGENT · MILESTONE #2

An assistant in **Slack** or **email** where lecturers request quizzes ("20 questions on lecture 4, mostly medium") and receive the drafts for review.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- No question reaches students until the lecturer has reviewed it.
- Questions are generated only from the lecturer's own material, not copied from textbooks or the web.
- Student scores and answers are visible only to the lecturer and the student.

SESSIONS IT DRAWS ON MOST

S22 AI in automations · S24 Multi-agent · S25 Files & OCR · S26 File generation & scheduling · S27 Validation & recovery · S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form Trigger · Google Gemini (document) · AI Agent + Agent Tools · Structured Output Parser · Loop Over Items · Google Sheets · Google Forms / HTTP Request · HTML to PDF · Gmail

SUCCESS CRITERIA

- A 20-question quiz from one lecture where a human checker finds at most 1 flawed question.
- Each question links back to the page it came from.

STRETCH GOAL

Auto-grading: read form responses, grade them and email each student personalised feedback on weak topics.

13 Real-Estate Listing Matcher

Real estate · Complexity ●●●●○

Collects property listings, structures them, and alerts buyers or renters the moment a listing matches what they asked for.

THE PROBLEM

Buyers and renters check many listing sites every day. Brokers and proptech startups want automated matching and instant alerts.

HOW IT WORKS · MILESTONE #1

1. Buyers register their preferences (area, budget, rooms, payment plan) through a form or the web chat agent.
2. A schedule collects listings from sources the team is allowed to use (public APIs, RSS, agency feeds or a provided dataset).
3. An extractor turns free-text listings (Arabic or English) into structured fields and drops incomplete or duplicate ones.
4. Listings are embedded so buyers can also search in natural language ("quiet 3-bedroom near a metro station").
5. Matching listings are sent as a daily digest or instant alert, with a short AI explanation of why each one fits.

DEPLOYED AGENT · MILESTONE #2

A property search agent on **web chat** ("quiet 3-bedroom near a metro station under 4M"), with daily match digests by **email**.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- Listings come only from sources whose terms allow it, and every listing links to its original source.
- Buyers are never filtered or ranked by religion, nationality or family status.
- Any AI price estimate is clearly labelled as an estimate, separate from the listed price.

SESSIONS IT DRAWS ON MOST

S17 APIs & webhooks | S21 Intermediate workflows | S23 RAG | S26 File generation & scheduling | S27 Validation & recovery | S30 Agent channels

SUGGESTED N8N NODES AND SERVICES

Form / Chat Trigger | Schedule Trigger | HTTP Request / RSS Read | Information Extractor | Remove Duplicates | Embeddings + Vector Store | Postgres / Sheets | Gmail

SUCCESS CRITERIA

- Matches a test set of 200 listings against 10 buyer profiles with clear match reasons.
- Respects each source's terms of use (documented in the report).

STRETCH GOAL

Price-fairness score: compare each listing with similar ones in the same area and flag over-priced units.

14 Talk-to-Your-Data Analyst Agent

Data & analytics · Complexity ●●●●●

Managers ask business questions in plain language; an agent writes safe SQL, runs it, and replies with numbers, a chart and an explanation.

THE PROBLEM

Non-technical staff depend on analysts for simple questions. Safe natural-language access to company data is a top request for AI automation engineers.

HOW IT WORKS · MILESTONE #1

1. A sample company database (sales, customers, products) is loaded into Postgres or Supabase.
2. Users ask questions via Slack or a web chat page; the agent has tools to read the schema and run **read-only** queries.
3. Generated SQL is checked by a guard step (SELECT only, row limit, allowed tables) before it runs; errors are fed back to the agent for a retry.
4. Results are summarised in plain language and, when useful, turned into a chart image (QuickChart API) attached to the reply.
5. A scheduled job sends a Monday KPI digest and logs every question with its SQL for auditing.

DEPLOYED AGENT · MILESTONE #2

A data analyst agent in **Slack** with memory, so managers can ask follow-ups ("and the same for last quarter?").

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The agent can only run read-only queries; any write or delete attempt is blocked and logged.
- Columns with personal data (names, phones, emails) are not available to the agent.
- Every answer shows the query and data period it used, so numbers can be checked.

SESSIONS IT DRAWS ON MOST

S22 AI in automations · S24 Multi-agent · S27 Validation & recovery · S29 Agent patterns & memory · S30 Agent channels · S33 Security & privacy

SUGGESTED N8N NODES AND SERVICES

Chat Trigger / Slack · AI Agent · Postgres Tool · Code (SQL guard) · HTTP Request (QuickChart) · Window Buffer Memory · Schedule Trigger · Postgres (audit log)

SUCCESS CRITERIA

- Answers 20 test business questions; at least 17 correct against analyst-verified answers.
- Any attempt to modify data is blocked and logged.

STRETCH GOAL

Anomaly alerts: a daily job detects unusual drops or spikes in key metrics and explains the likely cause.

15 IT Incident Response Automation

IT operations / DevOps · Complexity ●●●●●

Detects outages and errors, gathers diagnostics, opens a ticket with an AI first analysis, pages the on-call engineer and drafts the post-mortem.

THE PROBLEM

When systems fail, engineers lose the first minutes collecting logs and notifying people. Incident automation is a core skill for DevOps and platform teams.

HOW IT WORKS · MILESTONE #1

1. A schedule pings a list of websites/APIs; webhooks also receive alerts from monitoring tools or from n8n's own Error Trigger.
2. On failure, the workflow confirms with a second check to avoid false alarms, then collects recent logs and status details.
3. An agent classifies severity and writes a first analysis (likely cause, affected service, suggested next step).
4. A GitHub/Jira issue is opened and the on-call engineer is notified on Slack, escalating if not acknowledged within 10 minutes.
5. When the service recovers, the incident is closed and a post-mortem draft with a timeline is generated as a document.

DEPLOYED AGENT · MILESTONE #2

An incident agent in **Slack**: engineers ask "What changed before the outage?" or "Summarise incident #42" inside the incident thread.

EXAMPLE ETHICS GUARDRAILS · MILESTONE #3

- The AI never runs fixes or restarts on its own; every action needs an engineer's approval.
- Logs are scrubbed of passwords, tokens and customer data before they reach the model.
- If no human acknowledges a critical alert within 10 minutes, it escalates; it never silently closes.

SESSIONS IT DRAWS ON MOST

S17 APIs & webhooks | S21 Intermediate workflows | S22 AI in automations | S26 File generation & scheduling | S27 Validation & recovery | S32 Monitoring & cost

SUGGESTED N8N NODES AND SERVICES

Schedule Trigger | Webhook | HTTP Request | Error Trigger | AI Agent | GitHub / Jira | Slack | Wait | Google Docs / Convert to File

SUCCESS CRITERIA

- A live demo: stop a test service and show detection, ticket, alert, escalation and recovery.
- Repeated failures of the same service are grouped into one incident.

STRETCH GOAL

A status page served from an n8n webhook that shows current incidents and uptime for the last 30 days.